

1. If the standard deviation of a dataset is small, what does this tell you about the data points?

- (A) They are widely dispersed.
- (B) They are clustered closely around the mean.
- (C) The data is skewed.
- (D) The data is symmetrical.

2. Consider the small data set from the notes.

6 7 7 7 8 8 9 9 10 11 11

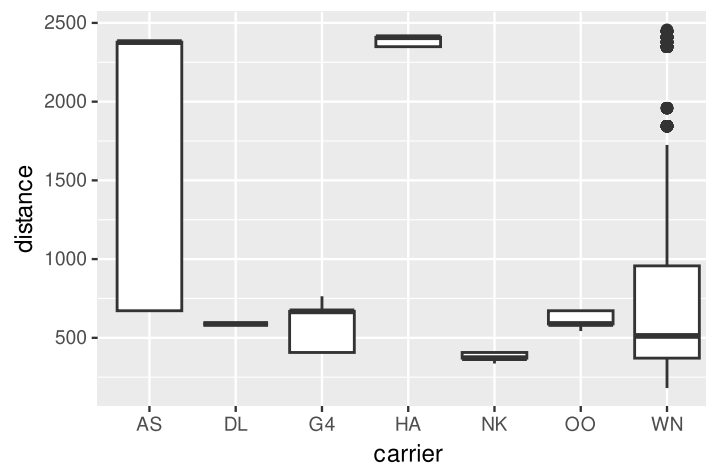
• If every value in this data set is increased by 10, what happens to the mean and median of the set?

- (A) Both remain the same.
- (B) Both increase by 10.
- (C) The mean increases by 10, but the median remains the same.
- (D) The mean remains the same, but the median increases by 10.

3. For which one of the following distributions will the median be a better measure of center than the mean?

- (A) Salary data for players in the National Basketball Association (NBA) where most of the players earn the league minimum and a few superstars earn very high salaries in comparison.
- (B) Repeated weight measurements of the same 1.6-ounce bag of Peanut M&Ms by students in a large chemistry class using a non-digital balance scale.
- (C) Height data from a large random sample of men.
- (D) Exam scores with a central peak around an average test score and a few students scoring lower than the average and a few scoring higher.

Below is a smaller version of the data from a future lab called `flights_mini`. It contains all flights out of Oakland (OAK) from December 2020. This data frame is used to create the plot that follows. `distance` refers to the distance a given plane travels on its flight, measured in miles. `carrier` refers to the carrier code for a specific airline.



4. Which of the following interpretations of the plot above are true? *Select all that apply.*

- (A) The carrier with the most heavily skewed distance distribution is HA.

- (B) The median distance of the flights operated by DL, G4, and OO are roughly equivalent.
- (C) The minimum distance traveled in this data set is roughly 200.
- (D) The carrier with the greatest variability in distance, as measured by the IQR, is AS.

5. Sketch your best sense of the distribution of the following variable(s). For each, please:
- Use a form of statistical graphic that emphasizes the important elements of the distribution.
 - Label the axes and provide plausible values for the tick marks.
 - Describe in words the shape of the distribution.
 - State which measure of center and spread would be most appropriate and approximate their values.

Make a note of any assumptions you're making in interpreting these variable names.

Number of body piercings among Stat 20 students

Scores on an easy quiz among Stat 20 students